Task 3: Networking Basics for Cyber Security

IP Address

An IP address (Internet Protocol address) is a unique numerical identifier assigned to every device connected to a network that uses the Internet Protocol for communication. It serves two main functions: identifying the device on the network and providing its location, enabling data to be sent and received correctly.

MAC Address

MAC address (Media Access Control address) is a unique hardware identifier assigned to a network interface card (NIC) by the manufacturer. It is a 48-bit hexadecimal number, typically written as 12 characters in pairs separated by colons or hyphens (e.g., 00:1A:2B:3C:4D:5E or 00-1A-2B-3C-4D-5E).

DNS

Domain Name System (DNS) is a hierarchical and distributed naming system that translates human-readable domain names into machine-readable Internet Protocol (IP) addresses, enabling users to access websites without memorizing numerical addresses.

TCP

TCP, or Transmission Control Protocol, is a core protocol in the Internet Protocol suite (TCP/IP) responsible for reliably transmitting data between devices over a network.

UDP

User Datagram Protocol (UDP) is a core communication protocol in the Internet Protocol (IP) suite, operating at the transport layer. It enables fast, connectionless data transmission by sending independent packets, called datagrams, without establishing a prior connection or guaranteeing delivery, order, or error correction.

Install Wireshark

Windows:

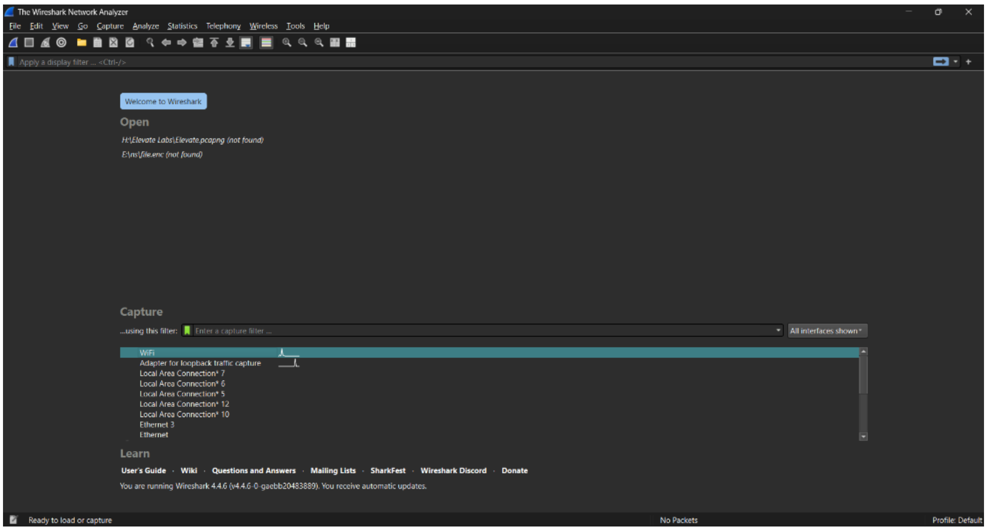
Download Wireshark → Install (select Npcap)

Linux:

sudo apt update

sudo apt install wireshark -y

Start Packet Capture (Live traffic)



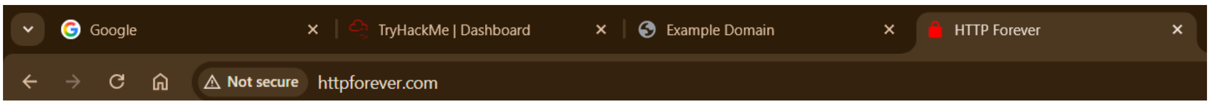
Select interface:

- Wi-Fi (if using Wi-Fi)
- Ethernet (LAN)
- In VM select: active adapter (NAT/Bridged)

Generate traffic for capture

Open browser and visit:

- google.com
- tryhackme.com
- example.com



Run CMD / Terminal commands:

Windows:

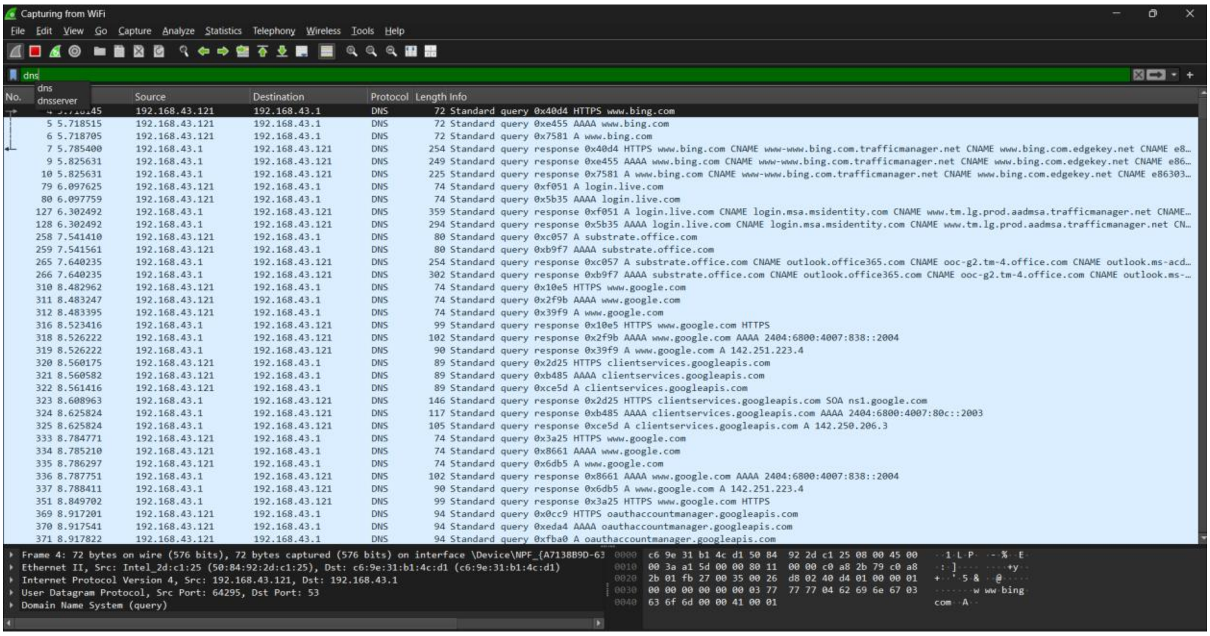
ping google.com

nslookup tryhackme.com

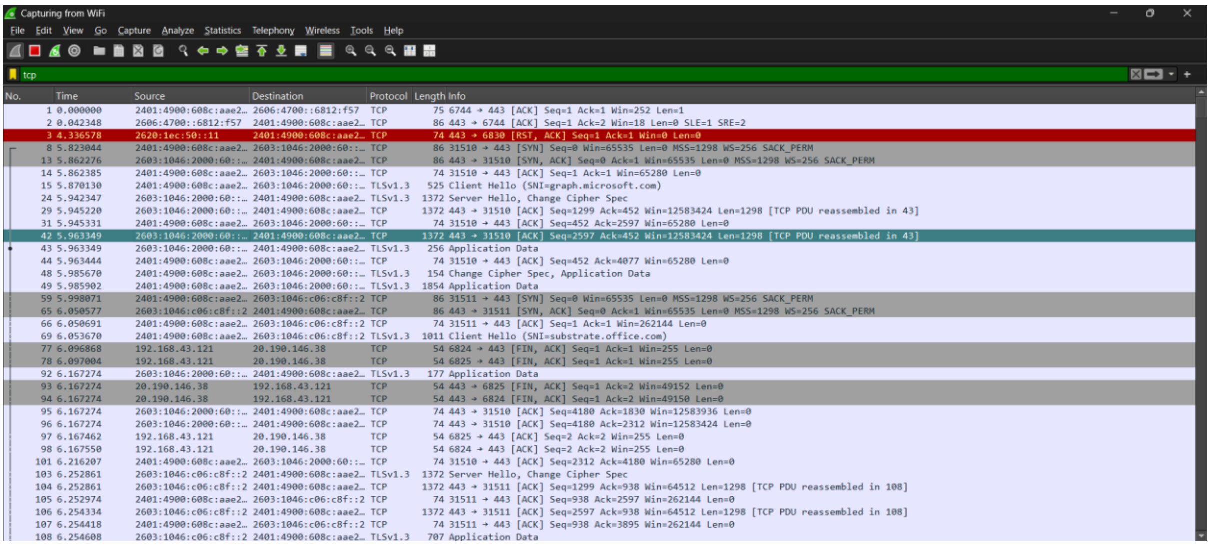
```
C:\Users\hari6>ping google.com

Pinging google.com [2404:6800:4007:83a::200e] with 32 bytes of data:
Reply from 2404:6800:4007:83a::200e: time=45ms
Reply from 2404:6800:4007:83a::200e: time=43ms
Reply from 2404:6800:4007:83a::200e: time=67ms
Reply from 2404:6800:4007:83a::200e: time=37ms

Ping statistics for 2404:6800:4007:83a::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 37ms, Maximum = 67ms, Average = 48ms

C:\Users\hari6>nslookup tryhackme.com
Server:  UnKnown
Address:  192.168.43.1

Non-authoritative answer:
Name:    tryhackme.com
Addresses:  2606:4700:83b6:d299:d33b:891:ae35:d89c
          172.66.164.239
          104.20.29.66
```

Filter packets by Protocol

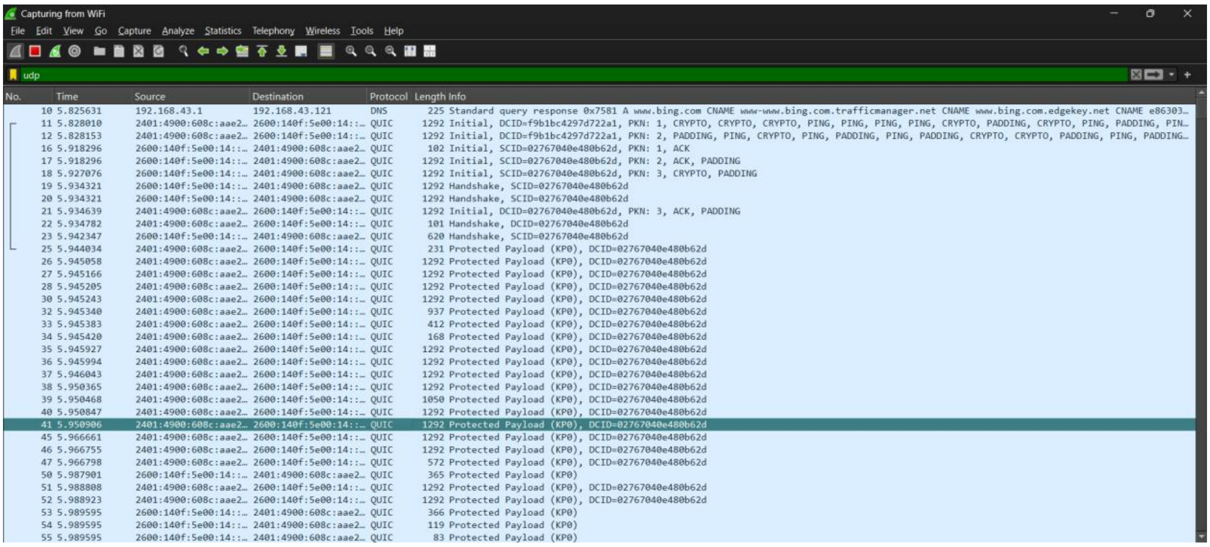
Show DNS only



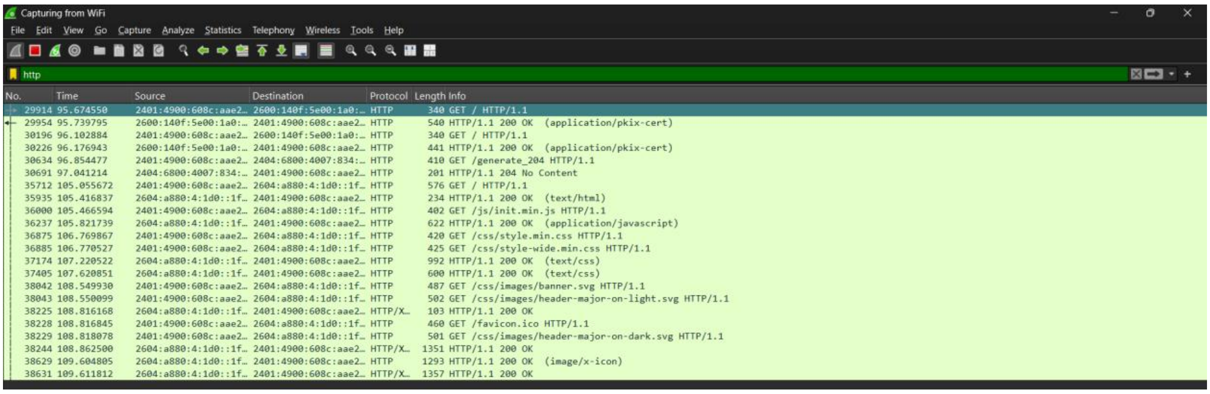
Show TCP only



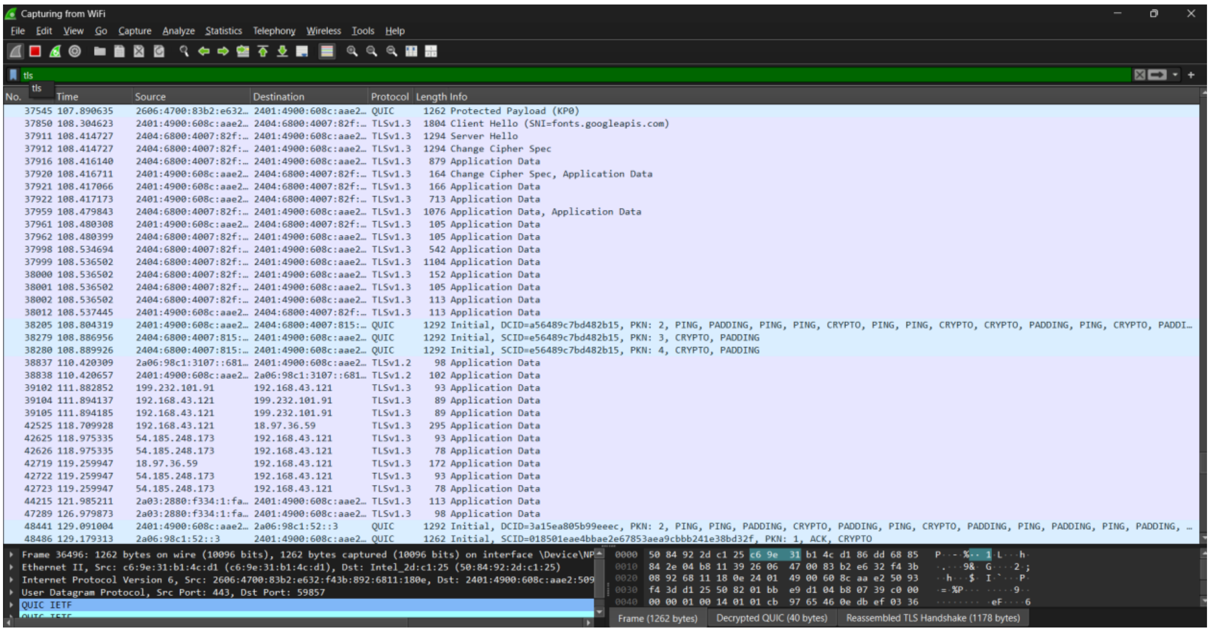
Show UDP only



Show HTTP (plain text)

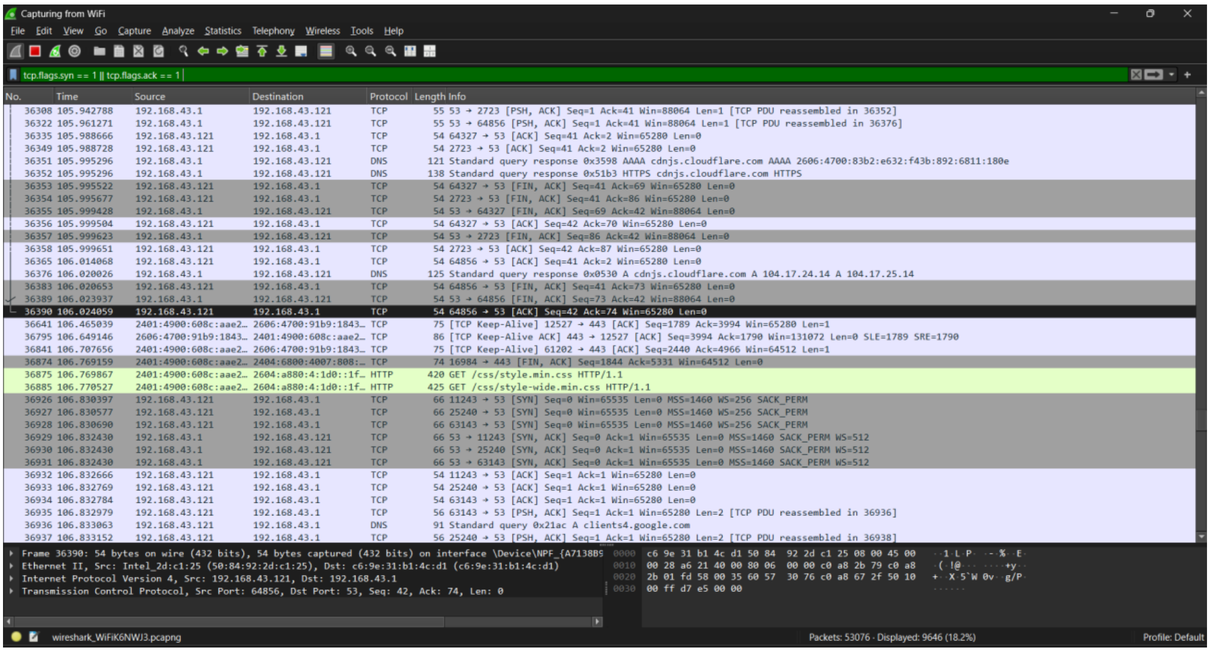


Show HTTPS (encrypted)



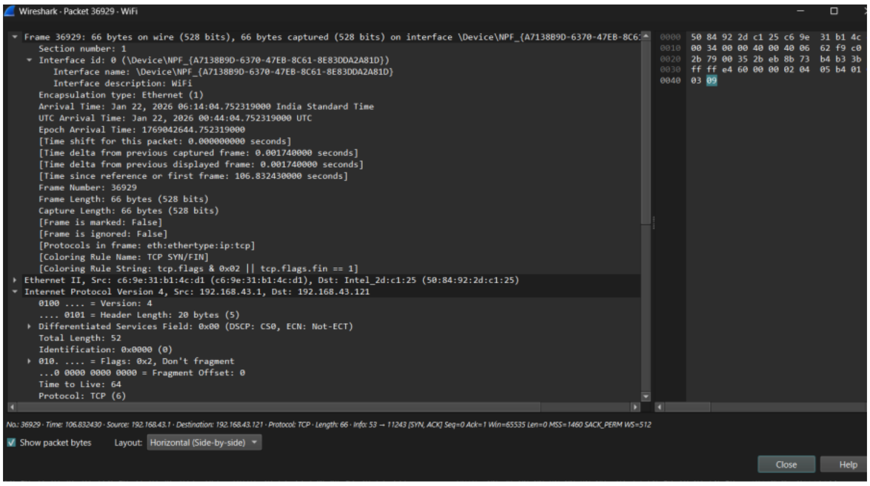
Observe TCP 3-way handshake

tcp.flags.syn == 1 || tcp.flags.ack == 1



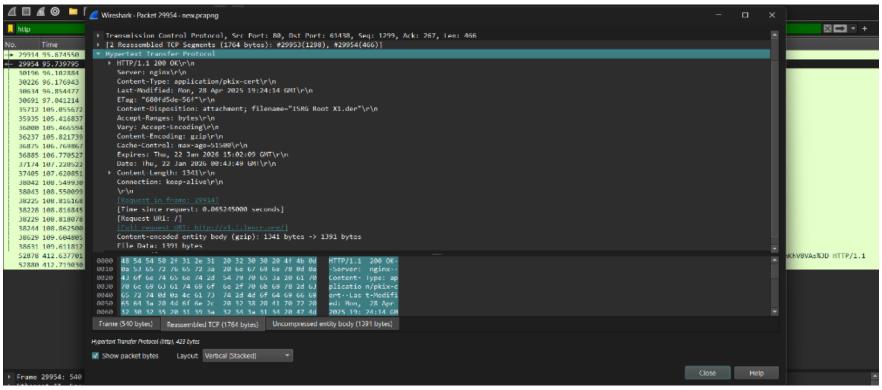
- SYN
- SYN-ACK
- ACK

Flags: SYN / ACK



Plain-text traffic vs Encrypted traffic

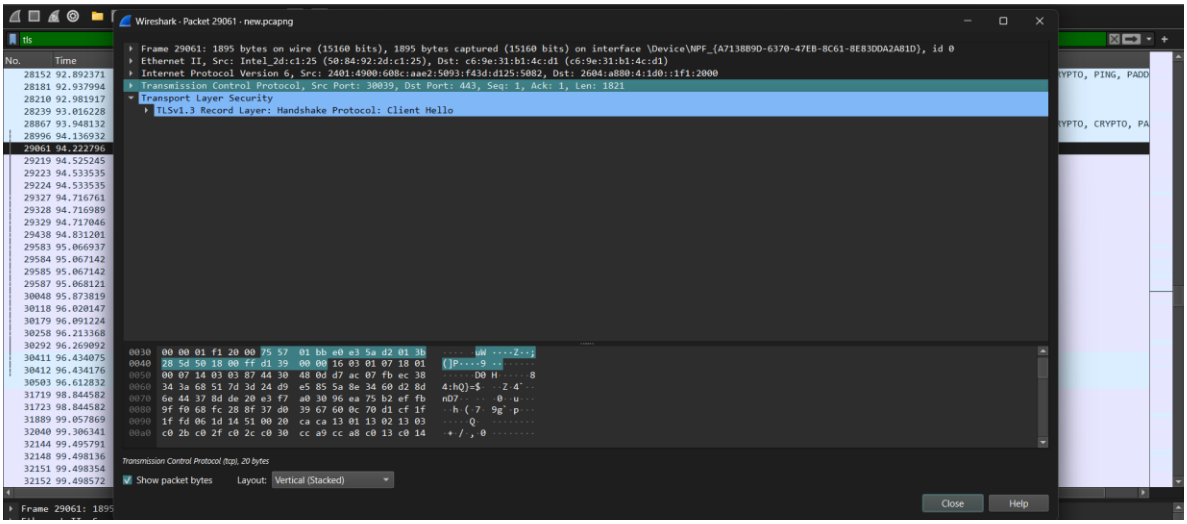
http

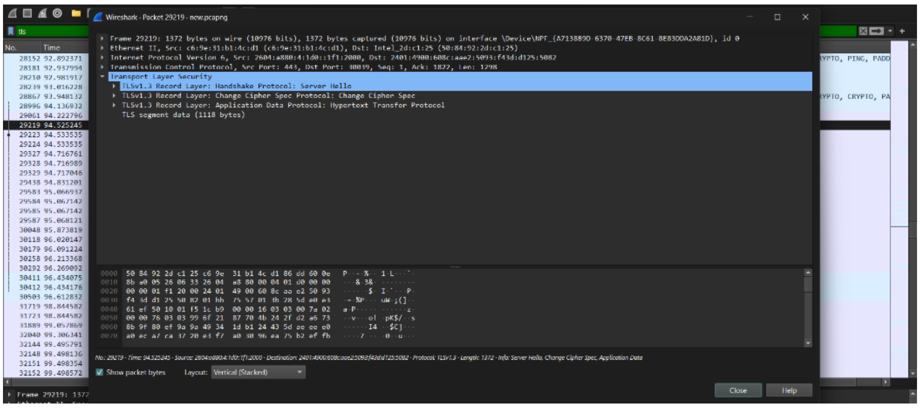


Tls

Click packet → you will see:

- Client Hello / Server Hello
- Encrypted Application Data





Capture DNS queries + analyze

Dns

- Standard query
- Standard query response

259	7.541561	192.168.43.121	192.168.43.1	DNS	80	Standard query 0xb9f7 AAAA substrate.office.com
265	7.640235	192.168.43.1	192.168.43.121	DNS	254	Standard query response 0xc057 A substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.tm-4.office.com CNAME outlook.ms-a
266	7.640235	192.168.43.1	192.168.43.121	DNS	302	Standard query response 0xb9f7 AAAA substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.tm-4.office.com CNAME outlook.m
310	8.482962	192.168.43.121	192.168.43.1	DNS	74	Standard query 0x10e5 HTTPS www.google.com
311	8.483247	192.168.43.121	192.168.43.1	DNS	74	Standard query 0x2f9b AAAA www.google.com
312	8.483395	192.168.43.121	192.168.43.1	DNS	74	Standard query 0x39f9 A www.google.com
316	8.523416	192.168.43.1	192.168.43.121	DNS	99	Standard query response 0x10e5 HTTPS www.google.com HTTPS
318	8.526222	192.168.43.1	192.168.43.121	DNS	102	Standard query response 0x2f9b AAAA www.google.com AAAA 2404:6800:4007:838::2004
319	8.526222	192.168.43.1	192.168.43.121	DNS	90	Standard query response 0x39f9 A www.google.com A 142.251.223.4
320	8.560175	192.168.43.121	192.168.43.1	DNS	89	Standard query 0x2d25 HTTPS clientservices.googleapis.com
321	8.560582	192.168.43.121	192.168.43.1	DNS	89	Standard query 0xb485 AAAA clientservices.googleapis.com
322	8.561416	192.168.43.121	192.168.43.1	DNS	89	Standard query 0xce5d A clientservices.googleapis.com
323	8.608963	192.168.43.1	192.168.43.121	DNS	146	Standard query response 0x2d25 HTTPS clientservices.googleapis.com SOA ns1.google.com
324	8.625824	192.168.43.1	192.168.43.121	DNS	117	Standard query response 0xb485 AAAA clientservices.googleapis.com AAAA 2404:6800:4007:80c::2003
325	8.625824	192.168.43.1	192.168.43.121	DNS	105	Standard query response 0xce5d A clientservices.googleapis.com A 142.250.206.3
333	8.784771	192.168.43.121	192.168.43.1	DNS	74	Standard query 0x3a25 HTTPS www.google.com
334	8.785210	192.168.43.121	192.168.43.1	DNS	74	Standard query 0x8661 AAAA www.google.com
335	8.786297	192.168.43.121	192.168.43.1	DNS	74	Standard query 0x6db5 A www.google.com
336	8.787751	192.168.43.1	192.168.43.121	DNS	102	Standard query response 0x8661 AAAA www.google.com AAAA 2404:6800:4007:838::2004
337	8.788411	192.168.43.1	192.168.43.121	DNS	90	Standard query response 0x6db5 A www.google.com A 142.251.223.4
351	8.849702	192.168.43.1	192.168.43.121	DNS	99	Standard query response 0x3a25 HTTPS www.google.com HTTPS
369	8.917201	192.168.43.121	192.168.43.1	DNS	94	Standard query 0x0cc9 HTTPS oauthaccountmanager.googleapis.com
370	8.917541	192.168.43.121	192.168.43.1	DNS	94	Standard query 0xeda4 AAAA oauthaccountmanager.googleapis.com
371	8.917822	192.168.43.121	192.168.43.1	DNS	94	Standard query 0xfba0 A oauthaccountmanager.googleapis.com